

Home-Work 4

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Problem 1 [10 pts]

Find the inverse Laplace transforms of the following functions:

(a) $\frac{(2s+5)e^{-2s}}{s^2+5s+6}$

$$X_1(s) = \frac{2s+5}{s^2+5s+6}$$

$$X_1(s) = \frac{2s+5}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

• by partial fractions.

for $A (s=-2)$

$$A = \frac{-4+5}{-2+3} = 1$$

for $B (s=-3)$

$$B = \frac{-6+5}{-3+2} = 1$$

so,

$$X_1(s) = \frac{1}{s+2} + \frac{1}{s+3}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at} u(t).$$

$$x_1(t) = (e^{-2t} + e^{-3t}) u(t)$$

• original function has a factor of e^{-2s} .

according to time shifting property $\mathcal{L}\{x(t-b)\} = e^{-sb} X(s)$

so,

$$x(t) = x_1(t-2) = (e^{-2(t-2)} + e^{-3(t-2)}) u(t-2)$$

(b) $\frac{se^{-3s}+2}{s^2+2s+2}$

$$X(s) = \frac{s}{s^2+2s+2} e^{-3s} + \frac{2}{s^2+2s+2}$$

so, (completing square method)

$$X_1(s) = \frac{s}{(s+1)^2+1}$$

$$X_1(s) = \frac{s+1-1}{(s+1)^2+1} = \frac{s+1}{(s+1)^2+1} - \frac{1}{(s+1)^2+1}$$

taking inverse Laplace:-

$$x_1(t) = e^{-t} \cos(t) u(t) - e^{-t} \sin(t) u(t)$$

taking inverse Laplace:-

$$x_1(t) = e^{-t} \cos(t) u(t) - e^{-t} \sin(t) u(t)$$

• Applying time shifting property ($k=3$)

$$x_1(t) = e^{-(t-3)} (\cos(t-3) - \sin(t-3)) u(t-3)$$

for second term:-

$$X_2(s) = \frac{2}{(s+1)^2 + 1} = 2 \left(\frac{1}{(s+1)^2 + 1} \right)$$

$$x_2(t) = 2 e^{-t} \sin(t) u(t)$$

$$\text{Total} = x_1(t) + x_2(t)$$

$$x(t) = e^{-(t-3)} (\cos(t-3) - \sin(t-3)) u(t-3) + 2 e^{-t} \sin(t) u(t)$$

Problem 2 [10 pts]

Consider a causal LTIC system described by the differential equation

$$2\ddot{y}(t) + 6\dot{y}(t) = \dot{x}(t) - 4x(t)$$

- Using transform-domain techniques, determine the ZIR $y_{zir}(t)$ if $y(0^-) = -3$.
- Using transform-domain techniques, determine the ZSR $y_{zsr}(t)$ to the input $x(t) = e\delta(t - \pi)$.

a) we know that for ZIR $x(t) = 0$

so,

$$2\ddot{y}(t) + 6\dot{y}(t) = 0$$

$$2 \frac{dy}{dt} + 6y(t) = 0$$

$$2sX(s) - 2y(0^-) + 6Y(s) = 0 \quad \because \mathcal{L}\left\{\frac{dy}{dt}\right\} = sY(s) - y(0^-)$$

$$Y(s) [2s+6] = -6$$

$$Y(s) = \frac{-6}{2(s+3)} = \frac{-3}{s+3}$$

$\therefore$ inverse Laplace:-

$$y_{zir}(t) = -3e^{-3t} u(t)$$

b) as its ZSR, so

$$X(s) = e \cdot e^{-\pi s} = e^{1-\pi s}$$

$$2sY(s) + 6Y(s) = sX(s) - 4X(s)$$

$$Y(s) [2s+6] = X(s) [s-4]$$

$$Y(s) = \frac{e^{-s\pi}}{2s+6} (s-4)$$

$$Y(s) = \frac{e^{-s\pi}}{2} \frac{s+3-3-4}{(s+3)}$$

$$Y(s) = \frac{e^{-s\pi}}{2} \left[\frac{\cancel{s+3}}{\cancel{s+3}} - \frac{7}{s+3} \right]$$

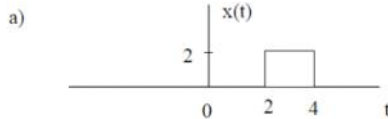
$$Y(s) = \frac{e^{-s\pi}}{2} - \frac{3.5e^{-s\pi}}{(s+3)}$$

$\therefore$ inverse transform:-

$$y_{\text{res}}(t) = \frac{e}{2} \delta(t-\pi) - 3.5 e^{-3(t-\pi)} u(t-\pi) \quad \text{Ans}$$

Problem 3 [10 pts]

Find the Fourier transform of the following signals. Sketch $|X(\omega)|$ in each case.



As it's a square function, its equation becomes:

$$x(t) = 2u(t-0) - 2u(t-4)$$

So,

$$X(\omega) = \int_0^4 2e^{-j\omega t} dt = 2 \left(\frac{e^{-j\omega t}}{-j\omega} \right) \Big|_0^4$$

$$= \frac{2}{-j\omega} (e^{-j4\omega} - e^{-j0\omega})$$

$$X(\omega) = \frac{2e^{-j2\omega}}{-j\omega} (e^{j2\omega} - e^{-j2\omega})$$

$$= \frac{2e^{-j2\omega}}{j\omega} (e^{j2\omega} - e^{-j2\omega}) \quad \therefore 2j\sin(\omega)$$

So,

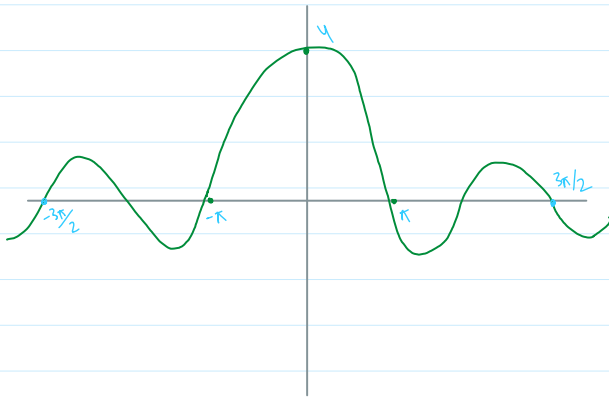
$$X(\omega) = \frac{2e^{-j2\omega}}{j\omega} \cdot 2j\sin(\omega) = \frac{4e^{-j2\omega}\sin(\omega)}{\omega}$$

So

$$h(\omega) = \frac{4e}{j\omega} \cdot \mathcal{L} \sin(\omega) = \frac{4e^{-}}{\omega} \sin(\omega)$$

so,

$$|X(\omega)| = 4|\text{sinc}(\omega)|$$

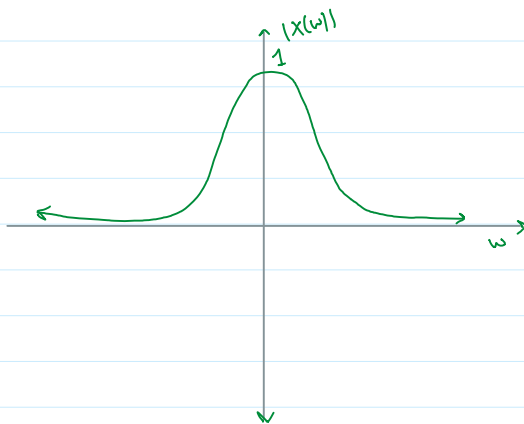


b) $x(t) = 2e^{-2t}u(t)$

$$e^{at}u(t) = \frac{1}{a+j\omega}$$

$$X(\omega) = \frac{2}{2+j\omega}$$

$$|X(\omega)| = \frac{2}{\sqrt{4+\omega^2}}$$



c) $x(t) = 5e^{-5t}u(t)$

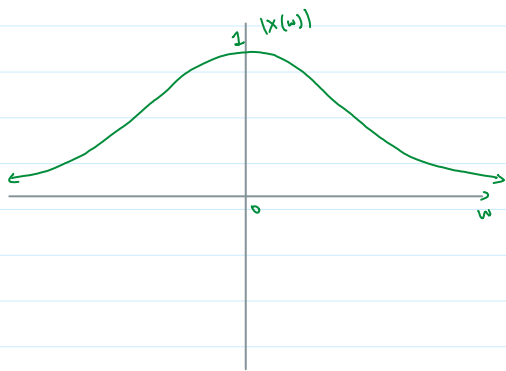
fourier transform will be:

$$X(\omega) = \frac{5}{5+j\omega}$$

so,

$$|X(\omega)| = \frac{5}{\sqrt{\omega^2 + 25}}$$

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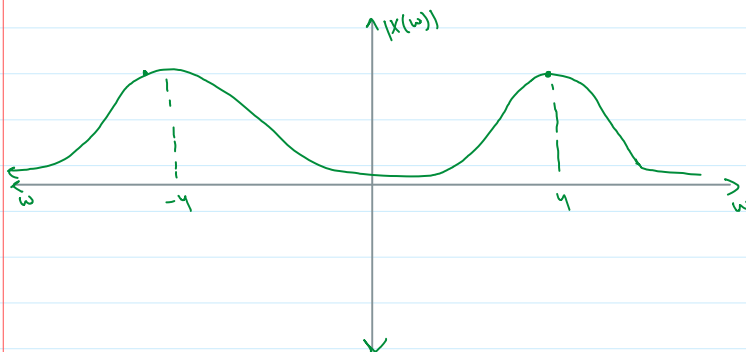
d) $x(t) = e^{-2t} \cos(4t) u(t)$

$$\therefore e^{-at} \cos(\omega_0 t) u(t) \leftrightarrow \frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$$

So,

$$X(\omega) = \frac{2 + j\omega}{(2 + j\omega)^2 + 16}$$

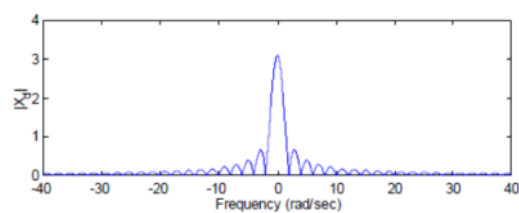
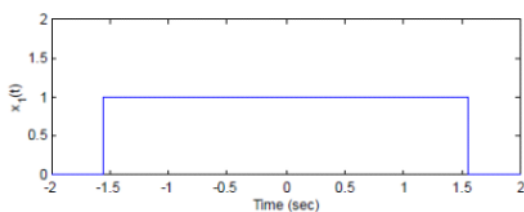
- This will now have two peaks nearby at $\omega = 4$ and $\omega = -4$.



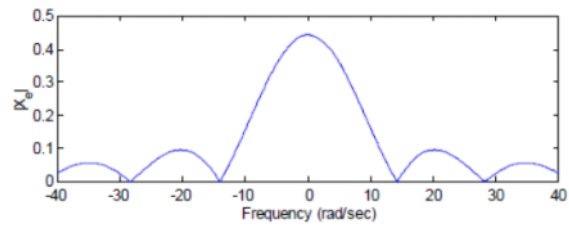
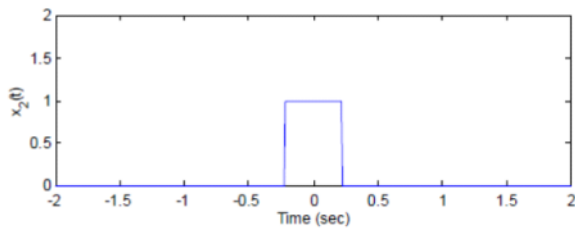
Problem 4 [10 pts]

Match the time responses with the corresponding frequency responses.

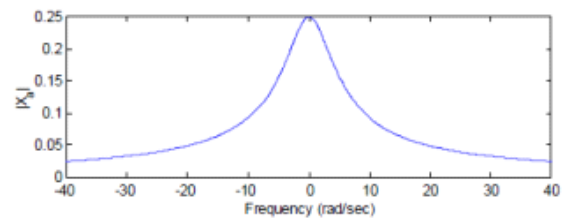
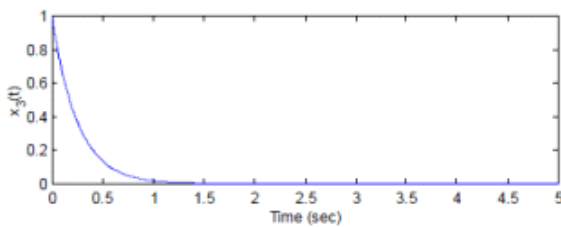
Justify your choices by providing a brief description.



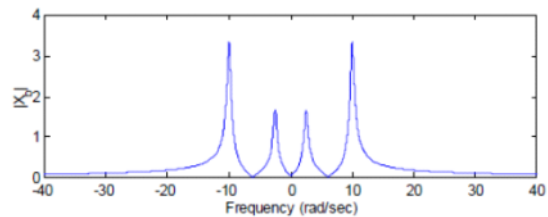
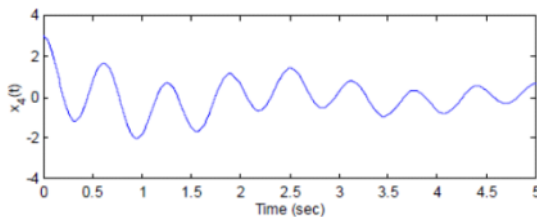
- Rectangular pulse in the time domain will always give sinc function.
- we observe that when we will make rectangle wide, it will shrink the function.



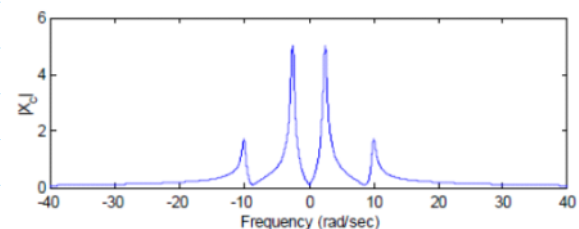
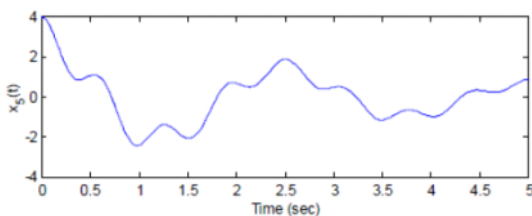
- as the rectangular pulse is now shrinked, so sinc function gets wider.



- as in time domain, the signal is decaying, it will give us a bell shaped graph in frequency domain.



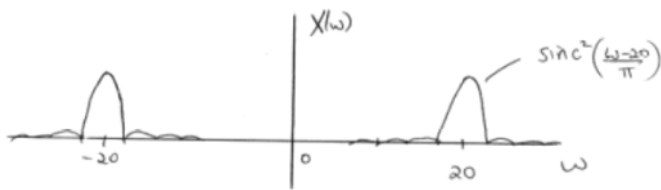
- In the time domain, the frequency of signal is higher, in frequency domain we can see that peaks are at higher frequency such as ± 10 rad/sec.



- as frequency of the signal is lower so main peaks are observed at lower ' ω ' in frequency domain.

Problem 5 [10 pts]

Compute the inverse Fourier transform of the following signal.



- The graph shows two sinc^2 lobes at $\omega = -20$ and $\omega = 20$.

$$\text{right one} = \text{sinc}^2\left(\frac{\omega - 20}{\pi}\right)$$

$$\text{left one} = \text{sinc}^2\left(\frac{\omega + 20}{\pi}\right)$$

so,

$$X(\omega) = \text{sinc}^2\left(\frac{\omega - 20}{\pi}\right) + \text{sinc}^2\left(\frac{\omega + 20}{\pi}\right)$$

we know that:

$$F\left\{\Delta\left(t/T\right)\right\} = T \text{sinc}^2\left(\frac{\omega T}{2}\right)$$

$$\therefore \frac{T}{2} = \frac{1}{\pi} \quad \text{so} \quad T = \frac{2}{\pi}$$

so

$$F\left\{\Delta\left(\frac{\pi t}{2}\right)\right\} = \frac{2}{\pi} \text{sinc}^2\left(\frac{\omega}{\pi}\right)$$

multiply both with $\pi/2$:

$$x_0(t) = F^{-1}\left\{\text{sinc}^2\left(\frac{\omega}{\pi}\right)\right\} = \frac{\pi}{2} \Delta\left(\frac{\pi t}{2}\right)$$

$$\therefore x(t) \cos(\omega_0 t) \Leftrightarrow \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$

$$X(\omega) = 2 \left[\frac{1}{2} X_0(\omega - 20) + \frac{1}{2} X_0(\omega + 20) \right]$$

Therefore,

time domain signal is:

$$x(t) = 2 x_0(t) \cos(20t)$$

Problem 6 [10 pts]

An **Exponential Fourier Series** is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Whereas, a **Trigonometric Fourier Series** is defined as

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

Find the relationship between the coefficients (c_k, a_0, a_k, b_k) of the two series.

- expanding the EFS:

$$x(t) = c_0 + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=-\infty}^{-1} c_k e^{jk\omega_0 t}$$

- Solving by changing third term k with $-k$ so,

$$x(t) = c_0 + \sum_{k=1}^{\infty} (c_k e^{jk\omega_0 t} + c_{-k} e^{-jk\omega_0 t})$$

$$e^{\pm jk\omega_0 t} = \cos(k\omega_0 t) \pm j \sin(k\omega_0 t)$$

now substitute:

$$\begin{aligned} x(t) &= c_0 + \sum_{k=1}^{\infty} [c_k (\cos k\omega_0 t + j \sin k\omega_0 t) + c_{-k} (\cos k\omega_0 t - j \sin k\omega_0 t)] \\ &= c_0 + \sum_{k=1}^{\infty} [(c_k + c_{-k}) \cos(k\omega_0 t) + j(c_k - c_{-k}) \sin(k\omega_0 t)] \end{aligned}$$

- Compare coefficients:-

$$DC = \frac{a_0}{2} = c_0 \quad a_0 = 2c_0$$

$$\text{Cosine : } a_k = c_k + c_{-k}$$

$$\text{Sine : } b_k = j(c_k - c_{-k})$$

so,

$$c_k = \frac{1}{2} (a_k - j b_k) \quad \& \quad c_{-k} = \frac{1}{2} (a_k + j b_k) \quad \text{Ans.}$$